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LightSpeed Plus/Ultra Axial Drive Belt Noise

christopher hardiman (christopher c hardiman), 05/16/2002 - 01:50 pm



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Communique Number

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Equipment

LightSpeed Plus and Ultra

Problem

Axial drive system is noisy during rotation. The axial drive belt should run/track in the center of the axial motor sprocket. A belt that is not tracking properly can cause excessive noise.

Solution

The most likely cause is that the motor mount bracket is skewed in relation to the main bearing. Please see attached document for details.

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Revision

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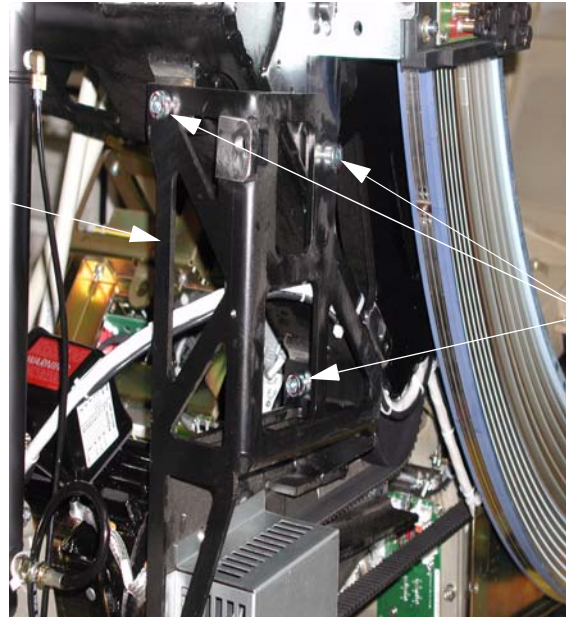
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LightSpeed Plus/Ultra Axial Drive Belt Noise

Problem: Axial drive system is noisy during rotation. The axial drive belt should run/track in the center of the axial motor sprocket. A belt that is not tracking properly can cause excessive noise.

Solution: The most likely cause is that the axial motor mount bracket is skewed in relation to the main bearing. See Figure 0-1. The axial motor mount bracket is attached to the system with three (3) bolts. The inside diameter of the bolt holes on the motor mount bracket is slightly larger than the diameter of the bolts resulting in enough play to cause tracking problems.

Axial Motor Mount
Bracket



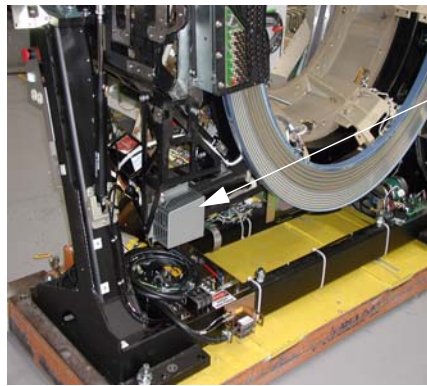
Mounting Bolts

Figure 0-1 Axial Motor Mount Bracket Bolts

If the belt is tracking to the rear of the motor drive gear, raising or “pulling up” on the front of the axial drive assembly will correct the skew and eliminate the noise.

If the belt is tracking towards the front of the motor drive gear, lowering or “pushing down” the front of the axial drive assembly will correct the skew and eliminate the noise.

- Procedure:**
- 1.) Refer to your System Service Manual as needed in the performance of these tasks.
 - 2.) Remove the side covers.
 - 3.) Remove both top covers.
 - 4.) Turn OFF the Axial Drive Enable Switch at the STC backplane.
 - 5.) Carefully remove the Mylar window.
 - 6.) Remove the rear cover.
 - 7.) Remove the axial drive pulley cover (See Figure 0-2).



Remove this cover.

Figure 0-2 Axial Drive Pulley Cover

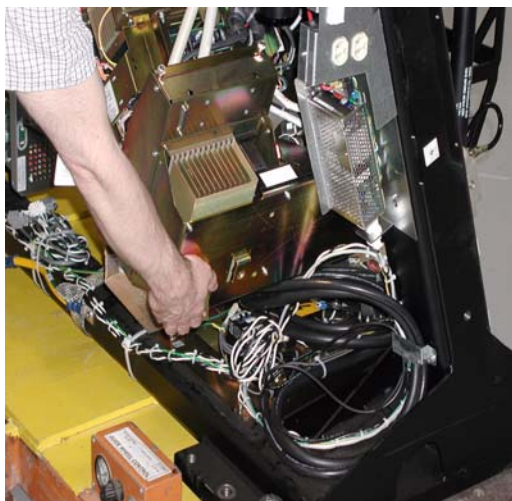
- 8.) At the STC backplane, turn ON the Axial Drive Enable.
- 9.) Under Axial Diagnostics on the console, initiate a 0.5s rotation for 10 seconds.
- 10.) Allow the gantry to spin up.
- 11.) Listen for the noise and, using a flashlight, observe the drive belt tracking.
Some Z-axis shifting of the belt is acceptable. The measure of an unacceptable position is the presence of noise and the observation that the belt is contacting the extreme edges of the drive pulley during operation.



DANGER

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- 12.) Allow the gantry to come to a complete stop.
- 13.) Turn OFF the Axial Drive Enable switch at the STC Backplane.
- 14.) Loosen, but leave snug, the 3 bolts securing the motor mount bracket to the stationary frame (See Figure 0-1).
- 15.) Work the axial drive assembly in the proper direction to correct the skew and then tighten the 3 mounting bolts to 66 +/- 3 N-m (48.7 +/- 2.2 lb.-ft.) (See Figure 0-3).



Shift the motor up or down to better align the axial drive motor mount weldment to the axial bearing.

Figure 0-3 Axial Motor Position Adjustment

- 16.) Repeat the procedure from step 8.) until the noise is eliminated and/or the belt tracks in the center of the drive pulley with minimal Z-axis motion.

- 17.) Allow the gantry to come to a complete stop.
- 18.) Turn OFF the Axial Drive Enable switch at the STC Backplane.
- 19.) If removed previously, reinstall the axial drive pulley cover. Bolt torque should be 8 N-m (71 inch-lbs, 6 lb.-ft.).
- 20.) One at a time, remove the three bolts securing the motor. Clean each of the bolt threads, apply Loctite 271 (p/n: 46-170684P2) and reinstall the bolts with a torque of 66 +/- 3 N-m (48.7 +/- 2.2 lb.-ft.).
- 21.) Reinstall the rear and top covers. Carefully reinstall the Mylar window.
- 22.) Turn ON the Axial Drive Enable switch at the STC Backplane.
- 23.) Reinstall the side covers.
- 24.) Perform a series of test scans to ensure proper system operation.

Comment: Thanks to Tom Neitzel for aiding in the development of this procedure and Seiji Uchida for identifying it.

